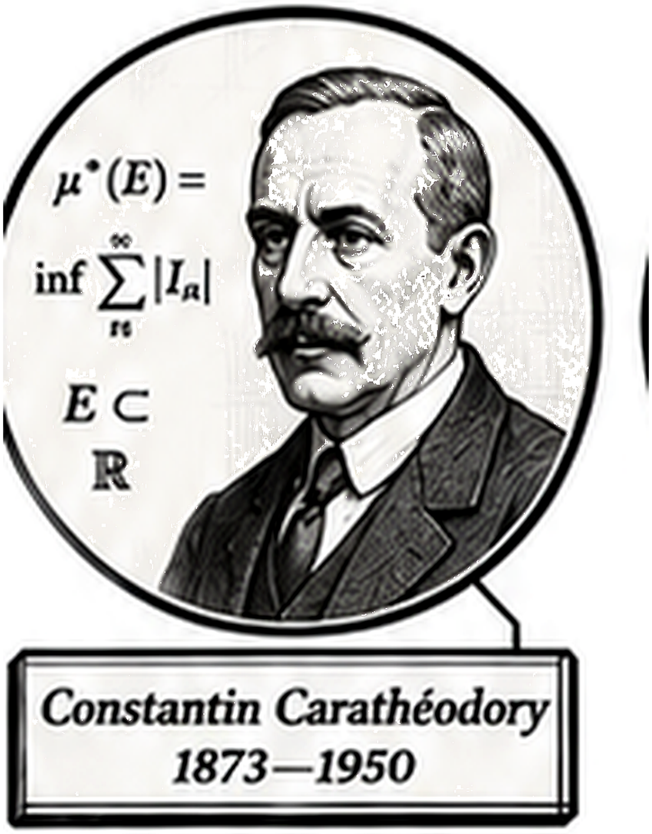
Volume IV

Mathematical Spaces



"I've got a number for you..."
— Constantin Caratheodory (probably)

Volume IV

Spaces

Self-Study Notes

William Sollers

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Part I

Spaces

Chapter 1

Metric Spaces

metric-spaces

Spaces $\rightarrow$ **Metric Spaces** $\rightarrow$ Topological Spaces

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Capstone Problem

Problem. TODO: state one synthesizing problem for Metric Spaces, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Chapter 2

Topology

topology

Topology and Geometry $\rightarrow$ **Topology** $\rightarrow$ Topology and Geometry

2.1 Topology

Remark (Status). This chapter is planned. Active mathematical content has not yet been added.

Proofs

Capstone Problem

Problem. TODO: state one synthesizing problem for Topology, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Capstone

Chapter 3

Set Algebra

set-algebra

Lattice Order → **Set Algebra** → Formalizing Algebraic Structures

3.1 Set Algebras

3.2 Systems of Sets

Comparison of Set Systems

Property	**Semiring**	**Ring**	**Algebra**	**σ-algebra**	**Topology**
Contains $\varnothing$	✓	✓	✓	✓	✓
Contains X	—	—	✓	✓	✓
Closed under $\cap$ (finite)	✓	✓	✓	✓	✓
Closed under $\cup$ (finite)	—	✓	✓	✓	✓
Closed under $\cup$ (countable)	—	—	—	✓	—
Closed under $\cup$ (arbitrary)	—	—	—	—	✓
Closed under complement	—	—	✓	✓	—
Difference $A \setminus B$	finite disj. union	✓	✓	✓	—

The definitions and theorems of the previous sections have been about individual sets and their operations. A second level of structure emerges when one asks: which *collections* of sets are closed under given operations? This question is not abstract generality for its own sake. The three subjects that depend most heavily on this chapter — topology, measure theory, and functional analysis — each begin by specifying a particular type of collection on a given set X. A topology specifies which subsets are "open." A σ-algebra specifies which subsets are "measurable." In both cases, the definition is precisely a list of closure conditions: which set operations applied to members of the collection must produce another member.

Mastering these four structures here, as purely set-theoretic objects, means that when topology and measure theory are encountered, all the foundational work is already done. The task will be to understand the *geometric* or *analytic* meaning of the axioms, not to struggle with the set-theoretic mechanics.

Definition (Semiring of Sets)

Definition 3.2.1 (Semiring of Sets)**.** A collection $\mathcal{S}$ of subsets of a set X is a *semiring* if:

(i) $\varnothing \in \mathcal{S}$;

(ii) $A, B \in \mathcal{S}$ implies $A \cap B \in \mathcal{S}$;

(iii) for any $A, B \in \mathcal{S}$, the difference $A \setminus B$ is a *finite disjoint union* of members of $\mathcal{S}$: there exist pairwise disjoint $C_1, \dots, C_n \in \mathcal{S}$ with $A \setminus B = C_1 \sqcup \cdots \sqcup C_n$.

Remark (Dependencies).

- Subset
- Empty Set
- Intersection
- Set Difference
- Union
- Pairwise Disjoint Family
- Finite and Infinite Sets

Example (Half-open intervals form a semiring). Let $X = \mathbb{R}$ and let $\mathcal{S}$ be the collection of all half-open intervals $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$, together with $\varnothing$. Then $\mathcal{S}$ is a semiring:

- $[a, b) \cap [c, d) = [\max(a, c), \min(b, d))$, which is again a half-open interval (or $\varnothing$);
- $[a, b) \setminus [c, d)$ is either $\varnothing$, $[a, c)$, $[d, b)$, or $[a, c) \sqcup [d, b)$ — in every case, a finite disjoint union of half-open intervals.

This semiring is the natural starting point for constructing Lebesgue measure: one first defines length on $\mathcal{S}$ (as $b - a$ for $[a, b)$), then extends to the generated ring and σ-algebra.

Remark (Why semirings appear in measure theory). Lebesgue measure is most naturally defined first on a semiring, because semirings have enough structure to state the countable additivity condition for a *premeasure*, yet are simple enough that explicit formulas for sets like $[a, b)$ are easy to write down. The general extension theorem (Carathéodory's theorem) then extends the premeasure to the full σ-algebra. This is why Kolmogorov & Fomin introduce semirings before algebras: they are the base case of the extension hierarchy.

Definition (Ring of Sets)

Definition 3.2.2 (Ring of Sets)**.** A collection $\mathcal{R}$ of subsets of X is a *ring of sets* if:

(i) $\varnothing \in \mathcal{R}$;

(ii) $A, B \in \mathcal{R}$ implies $A \cup B \in \mathcal{R}$;

(iii) $A, B \in \mathcal{R}$ implies $A \setminus B \in \mathcal{R}$.

Remark (Dependencies).

- Subset
- Empty Set
- Union
- Set Difference

Remark (Derived closure properties of a ring). A ring is closed under finitely many of each operation. Specifically, if $\mathcal{R}$ is a ring then:

- $A \cap B = A \setminus (A \setminus B) \in \mathcal{R}$ (intersection follows from set difference);
- $A \triangle B = (A \setminus B) \cup (B \setminus A) \in \mathcal{R}$ (symmetric difference follows from union and difference).

A ring is closed under all finite Boolean combinations of its elements.

Remark (Why "ring"). The name comes from algebra. Under the operations of symmetric difference $\triangle$ (playing the role of addition) and intersection $\cap$ (multiplication), a ring of sets is a commutative ring in the algebraic sense: $\triangle$ is commutative and associative with identity $\varnothing$, every element is its own inverse ($A \triangle A = \varnothing$), and $\cap$ distributes over $\triangle$.

Definition (Algebra of Sets)

Definition 3.2.3 (Algebra of Sets)**.** A collection $\mathcal{A}$ of subsets of X is an *algebra of sets* (or *Boolean algebra of sets*, or *field of sets*) if:

(i) $X \in \mathcal{A}$;

(ii) $A \in \mathcal{A}$ implies $A^c = X \setminus A \in \mathcal{A}$;

(iii) $A, B \in \mathcal{A}$ implies $A \cup B \in \mathcal{A}$.

Remark (Dependencies).

- Subset
- Complement
- Union

Remark (Derived properties of an algebra). An algebra is a ring that also contains X. Since $\varnothing = X^c$ and $X \in \mathcal{A}$, every algebra contains $\varnothing$. Closure under finite intersection follows: $A \cap B = (A^c \cup B^c)^c \in \mathcal{A}$ by De Morgan. An algebra is therefore closed under all finite Boolean operations: finite unions, finite intersections, complements, and differences.

Definition (σ-Algebra)

Definition 3.2.4 (σ-Algebra)**.** A collection $\mathcal{M}$ of subsets of X is a *σ-algebra* (or *σ-field*) on X if:

(i) $X \in \mathcal{M}$;

(ii) $A \in \mathcal{M}$ implies $A^c \in \mathcal{M}$;

(iii) if $\{A_n\}_{n\in\mathbb{N}}$ is a countable family in $\mathcal{M}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{M}$.

The pair $(X, \mathcal{M})$ is called a *measurable space*.

Remark (Dependencies).

- Subset
- Empty Set
- Complement
- Union
- Countable Set

Remark (Derived closure properties of a σ-algebra). Every σ-algebra is an algebra (take the union of a finite family by padding with $\varnothing$). But a σ-algebra is additionally closed under countable intersections: by De Morgan,

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c\right)^c \in \mathcal{M}.$$

The σ in "σ-algebra" is notation for "countable" (from the German *Summe*), signalling the step up from finite to countable closure.

Remark (Why countable and not arbitrary). A σ-algebra is closed under countable unions but not arbitrary ones. This is not a deficiency — it is the right condition for measure theory. Measure is defined to be countably additive: if $\{A_n\}$ is a countable disjoint family, then $\mu\big(\bigcup A_n\big) = \sum \mu(A_n)$. Extending to uncountable unions would force every set to have either measure zero or infinite measure (a useless theory). The countability restriction is what makes the theory nontrivial.

By contrast, a topology (defined below) is closed under *arbitrary* unions but only *finite* intersections. This is a different trade-off: open sets are defined by local conditions, and any union of open sets is open because being open is a pointwise property. Countable intersections of open sets are *not* required to be open; those are a separate class called G_δ sets.

Definition (Topology)

Definition 3.2.5 (Topology)**.** A *topology* on a set X is a collection τ of subsets of X satisfying:

(i) $\varnothing \in \tau$ and $X \in \tau$;

(ii) if $\{U_\alpha\}_{\alpha \in I}$ is any family of sets in τ (indexed by any set I), then $\bigcup_{\alpha \in I} U_\alpha \in \tau$;

(iii) if $U_1, \ldots, U_n \in \tau$ (finitely many), then $U_1 \cap \cdots \cap U_n \in \tau$.

The pair (X, τ) is a *topological space*. Members of τ are called *open sets*.

Remark (Dependencies).

- Subset
- Empty Set
- Indexed Family of Sets
- Indexed Union
- Indexed Intersection
- Finite and Infinite Sets

Remark (Why arbitrary unions but only finite intersections). A topology formalizes the idea of "nearness" without a distance function. The condition that arbitrary unions of open sets are open reflects the following: if a point x is near some set, and that set is a union of smaller sets, then x is near one of those smaller sets. This is a purely local condition and works for any union.

Finite intersections must be open because: if x lies in the interior of U_1 and also in the interior of U_2, it lies in the interior of $U_1 \cap U_2$. This argument works for finitely many sets (take the smallest neighborhood). For a countably infinite intersection, the "smallest neighborhood" might shrink to nothing — consider $\bigcap_{n=1}^{\infty}(-1/n, 1/n) = \{0\}$ in $\mathbb{R}$, which is not open. So infinite intersections of open sets are not required to be open.

Theorem (Intersection of σ-algebras is a σ-algebra)

Theorem 3.2.6 (Intersection of σ-algebras is a σ-algebra)**.** *Let $\{\mathcal{M}_\alpha\}_{\alpha \in I}$ be any family of σ-algebras on X (indexed by any set I). Then*

$$\mathcal{M} := \bigcap_{\alpha \in I} \mathcal{M}_\alpha$$

is a σ-algebra on X. *Go to proof.*

Remark (Standard quantified statement). $\big(\forall \alpha \in I,\ \mathcal{M}_\alpha$ is a σ-algebra on $X\big) \to \bigcap_{\alpha \in I} \mathcal{M}_\alpha$ is a σ-algeb
Explicitly: $X \in \bigcap \mathcal{M}_\alpha$; $A \in \bigcap \mathcal{M}_\alpha \to A^c \in \bigcap \mathcal{M}_\alpha$; $\forall n,\ A_n \in \bigcap \mathcal{M}_\alpha \to \bigcup_n A_n \in \bigcap \mathcal{M}_\alpha$.

Remark (Dependencies).

- σ-Algebra

- Indexed Family of Sets
- Intersection

Remark (Proof). Check each axiom: (i) $X \in \mathcal{M}_\alpha$ for all α, so $X \in \bigcap_\alpha \mathcal{M}_\alpha$. (ii) If $A \in \mathcal{M}$ then $A \in \mathcal{M}_\alpha$ for all α, so $A^c \in \mathcal{M}_\alpha$ for all α, so $A^c \in \mathcal{M}$. (iii) If $\{A_n\} \subseteq \mathcal{M}$ then $\{A_n\} \subseteq \mathcal{M}_\alpha$ for all α, so $\bigcup A_n \in \mathcal{M}_\alpha$ for all α, so $\bigcup A_n \in \mathcal{M}$. All three conditions are verified. The same argument works for algebras, rings, and topologies (with appropriate closure conditions).

Definition (Generated σ-Algebra)

Definition 3.2.7 (Generated σ-Algebra)**.** Let $\mathcal{F}$ be any collection of subsets of X. The *σ-algebra generated by $\mathcal{F}$*, written $\sigma(\mathcal{F})$, is the smallest σ-algebra on X that contains $\mathcal{F}$:

$$\sigma(\mathcal{F}) \;:=\; \bigcap \{\, \mathcal{M} \;\big|\; \mathcal{M} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{F} \subseteq \mathcal{M} \,\}.$$

The collection $\mathcal{F}$ is called a *generating family* for $\sigma(\mathcal{F})$.

Remark (Dependencies).

- σ-Algebra
- Subset
- Intersection
- Intersection of σ-Algebras

Remark (Why the definition is valid). The set of σ-algebras on X containing $\mathcal{F}$ is nonempty: $\mathcal{P}(X)$ is a σ-algebra containing every subset of X. Theorem 3.2.6 guarantees their intersection is again a σ-algebra. So $\sigma(\mathcal{F})$ is always well-defined. The same construction works for generated algebras, generated rings, and (with a small modification for topologies) generated topologies.

Remark (How to think about generated structures). $\sigma(\mathcal{F})$ contains $\mathcal{F}$, and then contains everything that must be present in any σ-algebra that contains $\mathcal{F}$: complements of elements of $\mathcal{F}$, countable unions of elements of $\mathcal{F}$, countable unions of complements of elements of $\mathcal{F}$, and so on — iterating indefinitely. The generated σ-algebra is the "closure" of $\mathcal{F}$ under all σ-algebraic operations. In practice, one specifies $\mathcal{F}$ as a convenient collection (e.g., intervals) and then works with the full σ-algebra by knowing which operations are allowed.

Definition (Borel σ-Algebra)

Definition 3.2.8 (Borel σ-Algebra)**.** Let (X, τ) be a topological space. The *Borel σ-algebra* on X, written $\mathcal{B}(X)$, is the σ-algebra generated by the open sets:

$$\mathcal{B}(X) \;:=\; \sigma(\tau).$$

Members of $\mathcal{B}(X)$ are called *Borel sets*.

Remark (Dependencies).

- Topology
- Generated σ-Algebra

Example (F_σ and G_δ sets). In a topological space X:

- A G_δ *set* is a countable intersection of open sets: $A = \bigcap_{n=1}^{\infty} U_n$ with each U_n open.
- An F_σ *set* is a countable union of closed sets: $A = \bigcup_{n=1}^{\infty} F_n$ with each F_n closed.

These names come from German: G for *Gebiet* (open set), δ for *Durchschnitt* (intersection); F for *Ferme* (closed), σ for *Summe* (union). Both G_δ and F_σ sets are Borel sets. They are dual: A is G_δ if and only if A^c is F_σ.

These classes appear in analysis and measure theory as the natural "next level" above open and closed sets. In $\mathbb{R}$: the set of continuity points of a monotone function is a G_δ set; the rational numbers $\mathbb{Q}$ are an F_σ set; the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ are a G_δ set.

Remark (Transition). The four structures — semiring, ring, algebra, σ-algebra — and the topology form a hierarchy of increasing closure strength. Measure theory builds from the bottom (semiring, ring) upward; topology is its own branch with a different closure pattern. The Borel σ-algebra bridges the two: it is generated by the topology, so it captures all sets describable by topological operations, and it is the standard "default" σ-algebra in any analytic context. Whenever a later chapter says "let f be a measurable function" without specifying the σ-algebra, it almost certainly means measurability with respect to the Borel σ-algebra.

3.3 The Power Set and Characteristic Functions

Proofs

Remark (Return). Return to Theorem

Theorem (Intersection of -algebras is a -algebra). *Let $\{\mathcal{M}_\alpha\}_{\alpha \in I}$ be any family of σ-algebras on X (indexed by any set I). Then*

$$\mathcal{M} := \bigcap_{\alpha \in I} \mathcal{M}_\alpha$$

is a σ-algebra on X.

Professional Standard Proof.
TODO: professional standard proof for sigma-intersection. □

Detailed Learning Proof.
TODO: detailed learning proof for sigma-intersection. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Capstone

Capstone Problem

Problem. TODO: state one synthesizing problem for Set Algebras, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Chapter 4

Algebras of Sets

algebras-of-sets

Lambda Calculus → **Algebras of Sets** → Torus Renders

4.1 Rings of Sets

Toolkit — Boolean Ring Examples

Concept family: Concrete Boolean ring examples and structural constraints.

Atomic items in this section:

- Remark: No Boolean ring of order 3 exists
- Example: The four-element Boolean ring $\mathbf{2}^2$
- Remark: Boolean rings have order 2^n

Key constraint: $(R, +)$ in any Boolean ring is an elementary abelian 2-group, so $|R| = 2^n$ for some $n \geq 0$.

Toolkit — Boolean Rings

Concept family: Boolean rings (Halmos–Givant §1) and their immediate consequences from the single idempotence axiom.

Atomic items in this section:

- Definition: Boolean Ring
- Definition: Idempotent Element
- Definition: Characteristic 2
- Lemma: Self Additive Inverse ($p + p = 0$)
- Lemma: Self Negation ($-p = p$)

Key predicate:

$$\mathrm{BooleanRing}(R, +, \cdot, -, 0, 1) \;:\equiv\; \mathrm{Ring}(R, +, \cdot, -, 0, 1) \;\wedge\; \mathrm{IdempotentRing}(R, \cdot).$$

Central theorem chain: Idempotence $\Rightarrow$ Characteristic 2 $\Rightarrow$ Self Negation ($-p = p$) $\Rightarrow$ Commutativity of $\cdot$.

Toolkit — Rings of Sets

Concept family: The ring axioms (Halmos–Givant §1) as the algebraic foundation for Boolean algebras of sets.

Atomic items in this section:

- Definition: Ring

Key predicate:

$$\mathrm{Ring}(R, +, \cdot, -, 0, 1) \;:\equiv\; \mathrm{AbelianGroup}(R, +, -, 0) \wedge \mathrm{Monoid}(R, \cdot, 1) \wedge \mathrm{Distributes}(\cdot, +, R).$$

Axiom numbering follows Halmos–Givant; axiom (4) (commutativity of multiplication) is not a ring axiom but a theorem in the Boolean setting.

Definition — Ring

Definition 4.1.1 (Ring)**.** A **ring** is a non-empty set R together with two binary operations $+$ and $\cdot$ on R, a unary operation $-$ on R, and two distinguished elements $0, 1 \in R$, satisfying the following axioms for all $p, q, r \in R$:

(1) $p + (q + r) = (p + q) + r$ *(additive associativity)*

(2) $p \cdot (q \cdot r) = (p \cdot q) \cdot r$ *(multiplicative associativity)*

(3) $p + q = q + p$ *(additive commutativity)*

(5) $p + 0 = p$ *(additive identity)*

(6) $p \cdot 1 = p$ *(multiplicative identity)*

(7) $p + (-p) = 0$ *(additive inverse)*

(8) $p \cdot (q + r) = p \cdot q + p \cdot r$ *(left distributivity)*

(9) $(q + r) \cdot p = q \cdot p + r \cdot p$ *(right distributivity)*

Axiom numbering follows Halmos–Givant Givant and Halmos [2009]; axiom (4) is commutativity of multiplication, which is not a ring axiom but a theorem in the Boolean setting. A ring possessing a distinguished element 1 satisfying axiom (6) is called a **ring with unit**. A ring satisfying $p \cdot q = q \cdot p$ for all $p, q \in R$ is called a **commutative ring**.

Remark (Standard quantified statement)*.*

$$
\begin{aligned}
(R, +, \cdot, -, 0, 1) \text{ is a ring } \iff \forall p, q, r \in R:&\\
&p + (q + r) = (p + q) + r\\
\wedge\quad &p \cdot (q \cdot r) = (p \cdot q) \cdot r\\
\wedge\quad &p + q = q + p\\
\wedge\quad &p + 0 = p\\
\wedge\quad &p \cdot 1 = p\\
\wedge\quad &p + (-p) = 0\\
\wedge\quad &p \cdot (q + r) = p \cdot q + p \cdot r\\
\wedge\quad &(q + r) \cdot p = q \cdot p + r \cdot p.
\end{aligned}
$$

Remark (Definition predicate reading)*.*

$$\mathrm{Ring}(R, +, \cdot, -, 0, 1) \iff \mathrm{AbelianGroup}(R, +, -, 0) \wedge \mathrm{Monoid}(R, \cdot, 1) \wedge \mathrm{Distributes}(\cdot, +, R)$$

Remark (Negated quantified statement).

$$
\begin{aligned}
(R,+,\cdot,-,0,1) \text{ is } \textbf{not} \text{ a ring} \iff \exists p,q,r \in R:\ & \\
& p+(q+r) \neq (p+q)+r \\
\lor \quad & p\cdot(q\cdot r) \neq (p\cdot q)\cdot r \\
\lor \quad & p+q \neq q+p \\
\lor \quad & p+0 \neq p \\
\lor \quad & p\cdot 1 \neq p \\
\lor \quad & p+(-p) \neq 0 \\
\lor \quad & p\cdot(q+r) \neq p\cdot q + p\cdot r \\
\lor \quad & (q+r)\cdot p \neq q\cdot p + r\cdot p.
\end{aligned}
$$

Remark (Failure modes)*.* The failure modes are exactly the ways that one of the defining structural conditions can fail.

Remark (Failure mode decomposition)*.* The eight axioms partition into three structural groups:

$$
\underbrace{(1),(3),(5),(7)}_{\text{AbelianGroup}(R,+,-,0)} \qquad \underbrace{(2),(6)}_{\text{Monoid}(R,\cdot,1)} \qquad \underbrace{(8),(9)}_{\text{Distributes}(\cdot,+,R)}
$$

A structure violating any axiom in the first group lacks a coherent additive group. A structure violating any axiom in the second group lacks a multiplicative identity or associativity. A structure violating the third group possesses two individually well-behaved operations that do not interact coherently; distributivity is the sole structural bridge between $+$ and $\cdot$, and its failure is sufficient to destroy the ring property.

Remark (Interpretation)*.* A ring abstracts the arithmetic of the integers. The additive structure is always a commutative group. The multiplicative structure is a monoid — associative with unit — but need not be commutative; axiom (4) of Halmos–Givant, commutativity of multiplication, is absent from the ring axioms. In the Boolean setting, multiplicative commutativity is not assumed but derived as a theorem from idempotence alone. Distributivity encodes the coherent interaction between the two operations: it is the axiom that makes $\cdot$ and $+$ form a single algebraic system rather than two independent ones.

Definition — Boolean Ring

Definition 4.1.2 (Boolean Ring)**.** A **Boolean ring** is an idempotent ring with unit. That is, a Boolean ring is a ring $(R,+,\cdot,-,0,1)$ satisfying the additional axiom:

$$
p\cdot p = p \qquad \text{for all } p \in R. \tag{11}
$$

An element p satisfying $p\cdot p = p$ is called **idempotent**. A ring in which every element is idempotent is called an **idempotent ring**.

The official axiom set for a Boolean ring consists of axioms (1)–(3), (5), (6), (8)–(11), where axiom (7) (the additive inverse law) is replaced by axiom (10) (characteristic 2), which is derived as a theorem. Axiom numbering follows Halmos–Givant Givant and Halmos [2009].

Remark (Standard quantified statement).

$$\mathrm{BooleanRing}(R, +, \cdot, -, 0, 1) \iff \mathrm{Ring}(R, +, \cdot, -, 0, 1) \ \wedge\ \forall p \in R\colon p \cdot p = p.$$

Remark (Definition predicate reading).

$$\mathrm{BooleanRing}(R, +, \cdot, -, 0, 1) \iff \mathrm{Ring}(R, +, \cdot, -, 0, 1) \ \wedge\ \mathrm{IdempotentRing}(R, \cdot)$$

Remark (Negated quantified statement).

$$\neg\ \mathrm{BooleanRing}(R, +, \cdot, -, 0, 1) \iff \neg\ \mathrm{Ring}(R, +, \cdot, -, 0, 1) \ \vee\ \exists p \in R\colon p \cdot p \neq p.$$

Remark (Failure modes). The failure modes are exactly the ways that one of the defining structural conditions can fail.

Remark (Failure mode decomposition). A structure fails to be a Boolean ring in exactly one of two ways:

$$\underbrace{\neg\ \mathrm{Ring}(R, +, \cdot, -, 0, 1)}_{\text{fails a ring axiom}} \qquad \vee \qquad \underbrace{\exists p \in R\colon p \cdot p \neq p}_{\text{some element is not idempotent}}$$

The second failure mode is the structurally interesting one. A ring in which even one element fails $p \cdot p = p$ is not Boolean. The simplest example: the ring $\mathbb{Z}$ of integers fails idempotence at every element except 0 and 1.

Remark (Canonical examples).

- $\mathbf{2} = \{0, 1\}$ with addition mod 2 and ordinary multiplication is the simplest non-degenerate Boolean ring.
- $\mathbf{2}^n$, the set of all n-tuples of elements of $\mathbf{2}$, with coordinatewise addition and multiplication, is a Boolean ring with 2^n elements. The zero is $(0, \ldots, 0)$ and the unit is $(1, \ldots, 1)$.
- $(\mathcal{P}(X), \triangle, \cap)$, the power set of any set X with symmetric difference as addition and intersection as multiplication, is a Boolean ring. Under the correspondence $A \leftrightarrow \mathbf{1}_A$ (the characteristic function of A), this ring is isomorphic to $\mathbf{2}^X$.

Remark (Interpretation). A Boolean ring is the minimal algebraic structure that forces the full Boolean calculus to emerge from a single additional axiom — idempotence of multiplication. The ring axioms supply the additive abelian group and the multiplicative monoid with distributivity. Idempotence then forces characteristic 2 (every element equals its own additive inverse), forces commutativity of multiplication, and identifies the ring with a field of sets via the representation theorem (Halmos–Givant, Chapter 22). The abstract definition thus captures exactly the algebra of set-theoretic operations: symmetric difference plays the role of addition and intersection plays the role of multiplication.

Remark (Dependencies).

- Definition: Ring

Example (The two-element Boolean ring). The simplest non-degenerate Boolean ring is the two-element ring $\mathbf{2} = \{0, 1\}$. Its addition and multiplication are given by the following tables.

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

Addition in **2** is addition modulo 2: $1+1=0$. Multiplication in **2** is ordinary multiplication of integers restricted to $\{0,1\}$. Every element of **2** satisfies $p+p=0$ and $p \cdot p = p$, making **2** the canonical example against which both axioms (10) and (11) are verified.

Definition — Idempotent Element

Definition 4.1.3 (Idempotent Element)**.** An element p of a ring R is called **idempotent** if

$$p \cdot p = p.$$

A ring in which every element is idempotent is called an **idempotent ring**.

Remark (Standard quantified statement)*.*

$$\text{IdempotentElement}(p, R, \cdot) \iff p \cdot p = p.$$
$$\text{IdempotentRing}(R, \cdot) \iff \forall p \in R\colon p \cdot p = p.$$

Remark (Definition predicate reading)*.*

$$\text{IdempotentElement}(p, R, \cdot) \qquad \text{IdempotentRing}(R, \cdot) \iff \forall p \in R\colon \text{IdempotentElement}(p, R, \cdot)$$

Remark (Negated quantified statement)*.*

$$\neg\, \text{IdempotentElement}(p, R, \cdot) \iff p \cdot p \neq p.$$

Remark (Interpretation)*.* Idempotence is the single additional axiom that forces Boolean structure. In the two-element ring **2**, verification is immediate: $0 \cdot 0 = 0$ and $1 \cdot 1 = 1$. In a power set algebra $\mathcal{P}(X)$, idempotence of intersection — $A \cap A = A$ — is the set-theoretic counterpart. The force of the axiom is not in these examples but in what it implies for an arbitrary ring: characteristic 2 and commutativity of multiplication both follow from idempotence alone, as the theorems below establish.

Remark (Dependencies)*.*

- Definition: Ring

Definition — Characteristic 2

Definition 4.1.4 (Characteristic 2)**.** A ring R is said to have **characteristic** 2 if

$$p + p = 0 \qquad \text{for all } p \in R. \tag{10}$$

Remark (Standard quantified statement)*.*

$$\text{Characteristic2}(R, +, 0) \iff \forall p \in R\colon p + p = 0.$$

Remark (Definition predicate reading)*.*

$$\text{Characteristic2}(R, +, 0) \iff \forall p \in R\colon \text{IdempotentElement}(p, R, +)$$

Remark (Negated quantified statement).

$$\neg\ \mathrm{Characteristic2}(R,+,0) \iff \exists p \in R\colon p+p \neq 0.$$

Remark (Interpretation). Characteristic 2 means that every element is its own additive inverse: $p + p = 0$ says precisely $p = -p$. The operation of negation therefore becomes the identity map — applying $-$ to any element returns that element unchanged. Halmos observes that in a ring of characteristic 2, the minus sign for additive inverses is never needed and is henceforth dropped. Axiom (7), $p + (-p) = 0$, is replaced in the official Boolean ring axiom set by the stronger equation (10), $p + p = 0$, which is a theorem derived from idempotence.

Remark (Dependencies).

- Definition: Boolean Ring
- Definition: Idempotent Element

Lemma

Lemma 4.1.5 (Self Additive Inverse)**.** *In any Boolean ring R,*

$$p + p = 0 \qquad \textit{for all } p \in R.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall p \in R\colon p + p = 0.$$

Remark (Definition predicate reading).

$$\mathrm{BooleanRing}(R,+,\cdot,-,0,1) \implies \mathrm{Characteristic2}(R,+,0).$$

Remark (Interpretation). This is Halmos consequence (a): idempotence of multiplication forces the additive structure to have characteristic 2. The proof expands $(p + q)^2$ by distributivity and idempotence, cancels p and q, then sets $p = q$ to obtain $p+p = 0$. The Lean 4 formalisation is `AdditiveIdempotence` in `BooleanAlgebraDefinitions.lean`.

Remark (Dependencies).

- Definition: Boolean Ring
- Definition: Characteristic 2

Lemma

Lemma 4.1.6 (Self Negation)**.** *In any Boolean ring R,*

$$-p = p \qquad \textit{for all } p \in R.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall p \in R\colon\ -p = p.$$

Remark (Interpretation). This follows immediately from 4.1.5. Since $p + p = 0$ and $p + (-p) = 0$ both hold, the uniqueness of additive inverses gives $-p = p$. The negation operation is therefore the identity map on any Boolean ring. Halmos uses this to drop the minus sign entirely from his notation for the rest of the book: since $-p = p$, writing $-p$ and writing p are the same thing.

Remark (Dependencies).

- Lemma: Self Additive Inverse

Remark (No Boolean ring of order 3). A Boolean ring of order 3 does not exist. The additive group $(R, +)$ of any Boolean ring satisfies $p + p = 0$ for all p, by Lemma 4.1.5. Every element therefore has additive order dividing 2, so $(R, +)$ is an elementary abelian 2-group. Elementary abelian 2-groups have order 2^n for some $n \geq 0$. Since 3 is not a power of 2, no Boolean ring of order 3 can exist.

Example (The four-element Boolean ring $\mathbf{2}^2$). The smallest Boolean ring with more than two elements is $\mathbf{2}^2 = \{(0,0),\ (0,1),\ (1,0),\ (1,1)\}$, the set of ordered pairs from $\mathbf{2} = \{0, 1\}$. Addition and multiplication are defined coordinatewise:

$$(p_0, p_1) + (q_0, q_1) \ = \ (p_0 + q_0,\ p_1 + q_1), \qquad (p_0, p_1) \cdot (q_0, q_1) \ = \ (p_0 \cdot q_0,\ p_1 \cdot q_1),$$

where each coordinate is computed in $\mathbf{2}$. The zero element is $(0, 0)$ and the unit is $(1, 1)$.

Addition table.

$+$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$(0,0)$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$(0,1)$	$(0,1)$	$(0,0)$	$(1,1)$	$(1,0)$
$(1,0)$	$(1,0)$	$(1,1)$	$(0,0)$	$(0,1)$
$(1,1)$	$(1,1)$	$(1,0)$	$(0,1)$	$(0,0)$

Multiplication table.

$\cdot$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$(0,0)$	$(0,0)$	$(0,0)$	$(0,0)$	$(0,0)$
$(0,1)$	$(0,0)$	$(0,1)$	$(0,0)$	$(0,1)$
$(1,0)$	$(0,0)$	$(0,0)$	$(1,0)$	$(1,0)$
$(1,1)$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$

Verification of idempotence. Every element squares to itself:

p	$p \cdot p$
$(0,0)$	$(0,0)$
$(0,1)$	$(0,1)$
$(1,0)$	$(1,0)$
$(1,1)$	$(1,1)$

Verification of characteristic 2**.** Every element is its own additive inverse:

p	$p+p$
$(0,0)$	$(0,0)$
$(0,1)$	$(0,0)$
$(1,0)$	$(0,0)$
$(1,1)$	$(0,0)$

Remark (Boolean rings have order 2^n). Every finite Boolean ring is isomorphic to $\mathbf{2}^n$ for some $n \geq 0$, and therefore has exactly 2^n elements. This follows from the representation theorem (Halmos–Givant, Chapter 22): every Boolean algebra is isomorphic to a field of sets, and every finite field of sets is isomorphic to $\mathcal{P}(X)$ for some finite set X with $|X| = n$, giving $|\mathcal{P}(X)| = 2^n$. The permissible finite sizes are $1, 2, 4, 8, 16, \ldots$ and no other cardinality is achievable.

Proofs

Remark (Return). Return to Lemma

Remark (Proof vault). Handwritten proof record

Lemma (Self Additive Inverse). *In any Boolean ring R,*

$$p + p = 0 \qquad \textit{for all } p \in R.$$

Professional Standard Proof.
Apply idempotence to the element $p + p$:

$$(p + p) \cdot (p + p) = p + p. \tag{i}$$

Expand the left side using right distributivity (9), then left distributivity (8), then collapse each $p \cdot p = p$ by idempotence (11):

$$(p+p)\cdot(p+p) \overset{(9)}{=} p\cdot(p+p)+p\cdot(p+p) \overset{(8)}{=} (p^2+p^2)+(p^2+p^2) \overset{p^2=p}{=} (p+p)+(p+p). \tag{ii}$$

Combining (i) and (ii): $(p+p)+(p+p) = p+p$. Setting $s = p+p$, the equation $s+s = s$ gives $s = 0$ by adding $-s$ and applying associativity, the inverse law, and the identity law. Hence $p + p = 0$. □

Detailed Learning Proof.

Step 1. Apply idempotence to the element $p+p$. The element $p+p$ is a member of the carrier of R. Axiom (11) (multiplicative idempotence) asserts every element of R is its own square. Apply this to the element $p + p$:

$$(p + p) \cdot (p + p) = p + p. \tag{i}$$

This is the pivot of the proof. It says $p + p$ squares to itself. Nothing about $p + p = 0$ has been assumed; that is the conclusion to be derived.

Step 2. Expand $(p + p) \cdot (p + p)$ using right distributivity. Axiom (9) (right distributivity) states:

$$(q + r) \cdot t = q \cdot t + r \cdot t \qquad \forall\, q, r, t \in R.$$

Set $q = p$, $r = p$, $t = p + p$:

$$(p + p) \cdot (p + p) = p \cdot (p + p) \;+\; p \cdot (p + p). \tag{ii}$$

The product has split into two identical copies of $p \cdot (p + p)$. This step holds in any ring; no idempotence has been used.

Step 3. Expand each $p \cdot (p + p)$ using left distributivity. Axiom (8) (left distributivity) states:

$$t \cdot (q + r) = t \cdot q + t \cdot r \qquad \forall\, t, q, r \in R.$$

Apply to $p \cdot (p + p)$ with $t = p$, $q = p$, $r = p$:

$$p \cdot (p + p) = p \cdot p \;+\; p \cdot p.$$

Apply to both copies in (ii):

$$p \cdot (p + p) + p \cdot (p + p) = (p \cdot p + p \cdot p) + (p \cdot p + p \cdot p). \tag{iii}$$

The right side contains four occurrences of $p \cdot p$. This expansion holds in any ring; idempotence has still not been used.

Step 4. Collapse each $p \cdot p$ to p by idempotence. Axiom (11) gives $p \cdot p = p$. Substitute into all four occurrences in (iii):

$$(p \cdot p + p \cdot p) + (p \cdot p + p \cdot p) = (p + p) + (p + p). \tag{iv}$$

This is the second and final use of idempotence.

Step 5. Chain (i) through (iv). Equations (i), (ii), (iii), (iv) combine as:

$$p + p \;=\; (p + p) \cdot (p + p) \;=\; (p + p) + (p + p).$$

Hence

$$(p + p) + (p + p) = p + p. \tag{v}$$

Step 6. Conclude $p + p = 0$ by cancellation. Let $s = p + p$. Equation (v) states $s + s = s$. Add the additive inverse $-s$ to the right of both sides, then apply associativity, the inverse law, and the identity law:

$$s+s = s \implies (s+s)+(-s) = s+(-s) \implies s+\underbrace{(s + (-s))}_{=0} = 0 \implies s+0 = 0 \implies s = 0.$$

Since $s = p + p$, the conclusion is $p + p = 0$. □

Remark (Proof structure). The proof is direct. The strategy applies idempotence twice: once to $p + p$ as a whole (Step 1) and once to p individually (Step 4), with distributivity (Steps 2–3) connecting the two applications. The conclusion follows by a standard additive cancellation (Step 6) using only the group axioms for $(R, +)$. No circularity arises: the proof never assumes $1 + 1 = 0$ or any consequence of characteristic 2. Both $p + p = 0$ and $1 + 1 = 0$ are consequences of idempotence; this proof derives the former directly without appealing to the latter. The Lean 4 formalisation is `AdditiveIdempotence` in `LRA/VolumeI/BooleanAlgebra/BooleanAlgebraDefinitions.lean`.

Remark (Dependencies).

- Definition of Ring — axioms (1) associativity, (5) identity, (7) inverse, (8) left distributivity, (9) right distributivity.
- Definition of Boolean Ring — axiom (11) multiplicative idempotence, applied in Steps 1 and 4.
- Definition of Idempotent Element — names the property $p \cdot p = p$.

Remark (Return). Return to Lemma

Remark (Proof vault). Handwritten proof record

Lemma (Self Negation). *In any Boolean ring R,*

$$-p = p \qquad \textit{for all } p \in R.$$

Professional Standard Proof.
By Self Additive Inverse, $p+p=0$. By the additive inverse axiom (7), $p+(-p)=0$. Since both p and $-p$ satisfy the equation $p+x=0$, and additive inverses are unique in any group (a standard group theory fact applied to the abelian group $(R,+)$), we conclude $-p=p$. □

Detailed Learning Proof.
Step 1. Recall the two equations that both hold in R.
From Self Additive Inverse (Lemma 4.1.5):

$$p + p = 0. \tag{i}$$

From the additive inverse axiom (7) of the ring definition:

$$p + (-p) = 0. \tag{ii}$$

Step 2. Apply uniqueness of additive inverses.
The additive group $(R, +, 0)$ is a group (abelian, by axiom (3)). In any group, if $a + b = 0$ and $a + c = 0$, then:

$$b = 0 + b = (a + c) + b \overset{\text{assoc}}{=} a + (c + b) \overset{\text{comm}}{=} a + (b + c) \overset{\text{assoc}}{=} (a + b) + c = 0 + c = c.$$

Apply with $a = p$, $b = p$ (from (i)) and $c = -p$ (from (ii)): both p and $-p$ are additive inverses of p, so $p = -p$. □

Remark (Proof structure). A one-step uniqueness argument. Self Additive Inverse shows p is its own additive inverse; the ring axiom asserts $-p$ is also an additive inverse of p; uniqueness of inverses in the underlying group forces $p = -p$. The proof has no calculation content — it is purely structural.

Remark (Dependencies).

- Lemma: Self Additive Inverse
- Definition: Ring (axiom (7), additive inverse law)

Capstone

Capstone Problem

Problem. TODO: state one synthesizing problem for Algebras Of Sets, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Chapter 5

Measure Spaces

measure-spaces

Topological Spaces → **Measure Spaces** → Modern Analysis

Proofs

Capstone

Capstone Problem

Problem. TODO: state one synthesizing problem for Measure Spaces, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Bibliography

Steven Givant and Paul R. Halmos. *Introduction to Boolean Algebras.* Undergraduate Texts in Mathematics. Springer, New York, 2009. ISBN 978-0-387-68436-9.